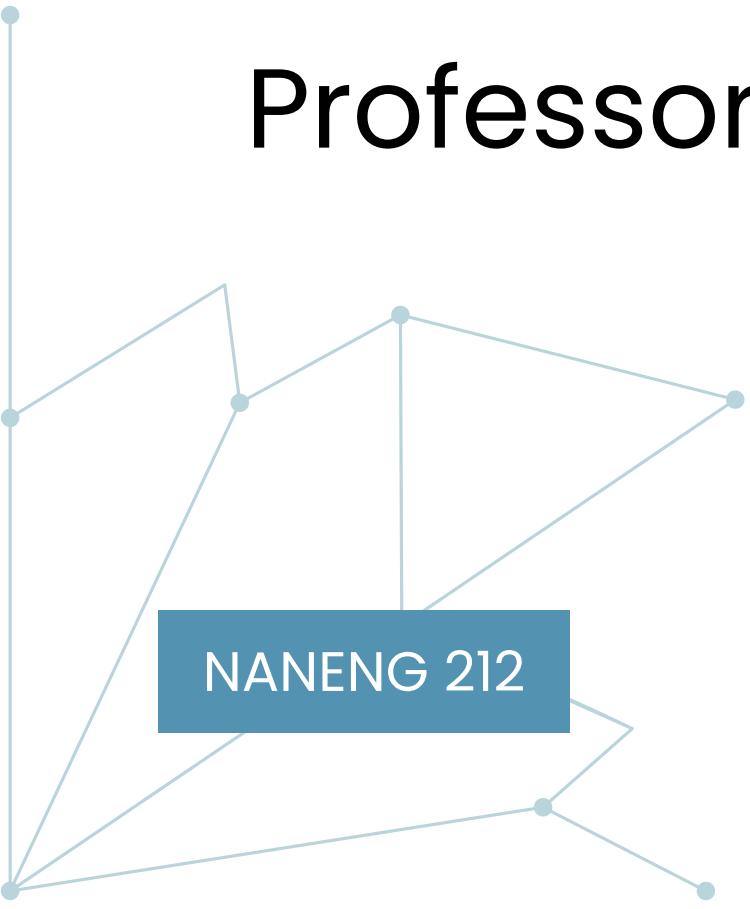


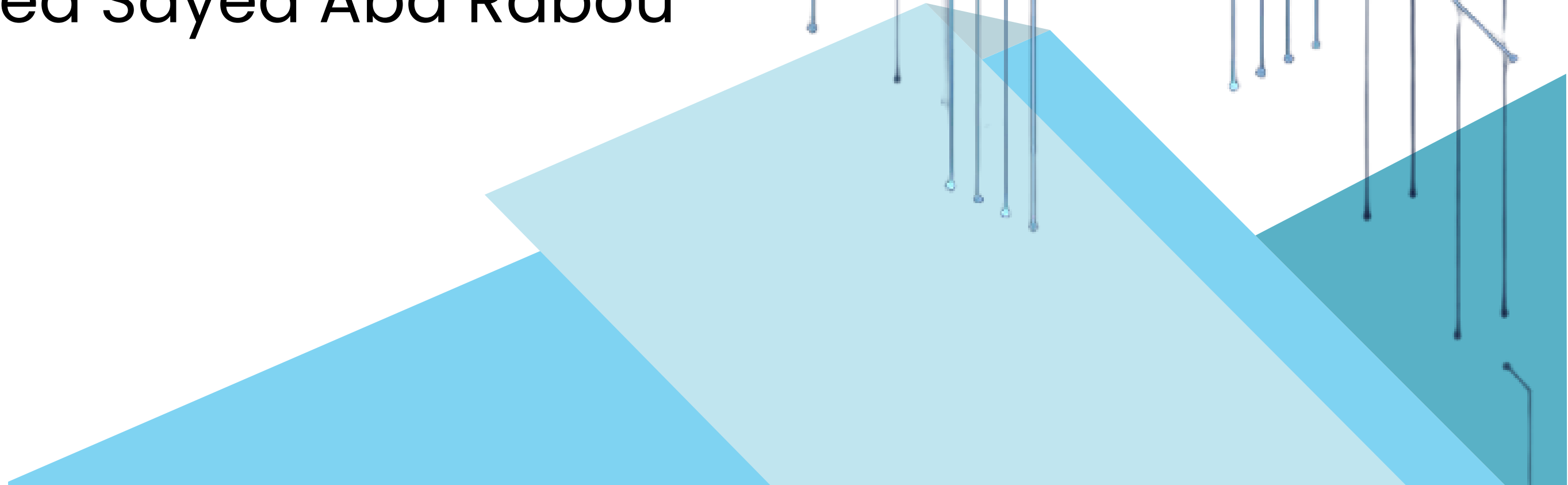
Self Balance Robot

■ Applied Digital Control

Professor Ahmed Sayed Abd Rabou



NANENG 212



State Space Model

A =

	x1	x2	x3	x4
x1	-0.02403	21.54	0.3202	-4.391
x2	-22.62	-0.7709	-26.58	18.26
x3	0.3578	25.14	-2.221	24.57
x4	-1.37	-1.421	2.538	-24.05

C =

	x1	x2	x3	x4
y1	-167.2	-17.36	4.039	0.6868

B =

	u1
x1	-0.0006114
x2	0.01898
x3	-0.00776
x4	0.005432

Continuous-time identified state-space model:

$$\frac{dx}{dt} = A x(t) + B u(t) + K e(t)$$

$$y(t) = C x(t) + D u(t) + e(t)$$

State-space Model Identification

Estimation data: Time domain data z

ESTIMATION PROGRESS

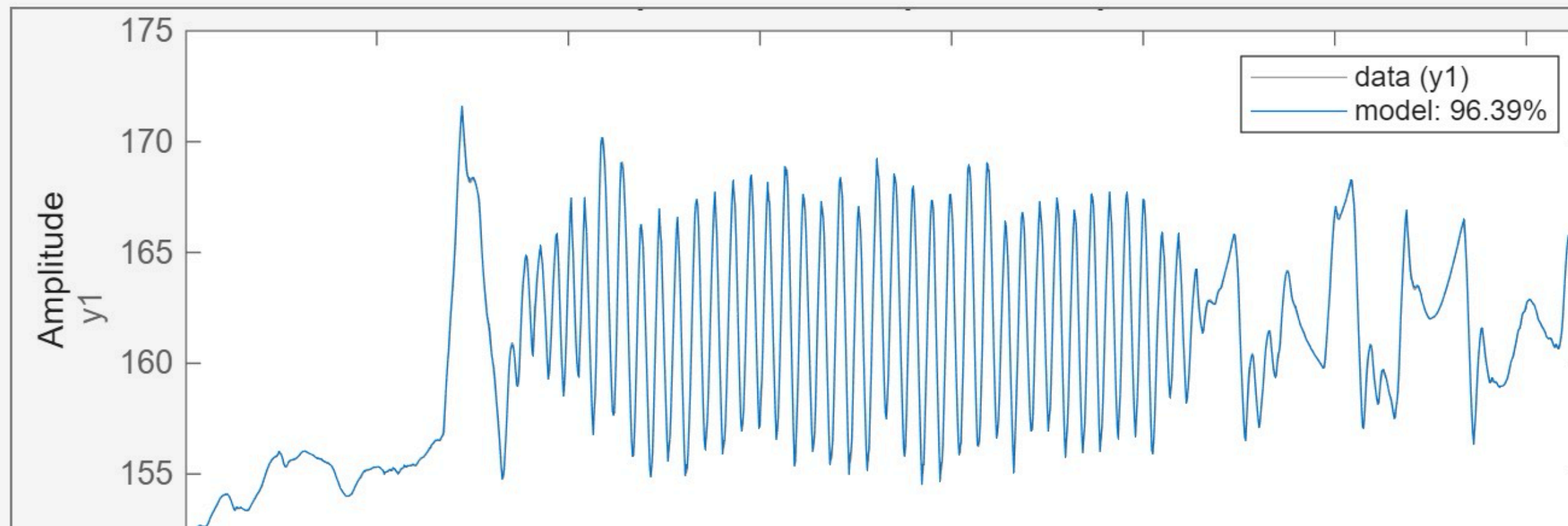
Estimating parameters using subspace algorithm...

Using N4Weight = "CVA"

Estimating parameter covariance...
done.

RESULT

Status: Estimated using N4SID with prediction focus
Fit to estimation data: **96.39%**, FPE: 0.0247903



Transfer function with K

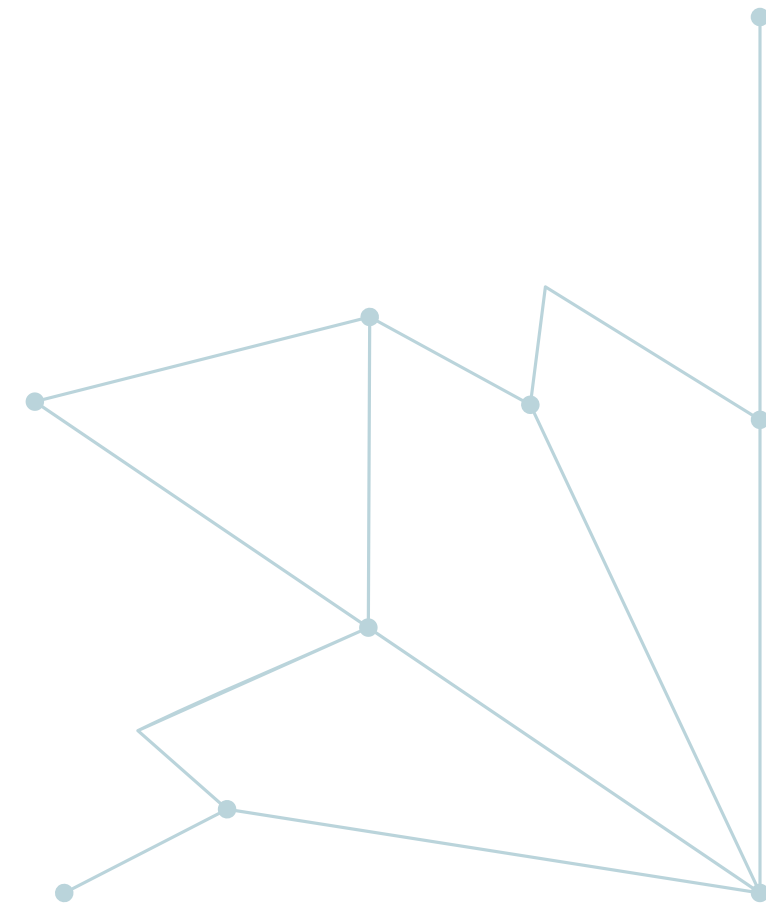
G_cl_s =

$$\frac{-0.2549 s^3 - 73.73 s^2 - 2859 s - 10.08}{s^4 + 26.99 s^3 + 1187 s^2 + 2.794e04 s - 2.329e04}$$

Transfer function without K

G_cl_s2 =

$$\frac{-0.2549 s^3 - 73.73 s^2 - 2859 s - 10.08}{s^4 + 27.07 s^3 + 1187 s^2 + 2.791e04 s - 2.319e04}$$



poles With K vs Without k

```
ans =
```

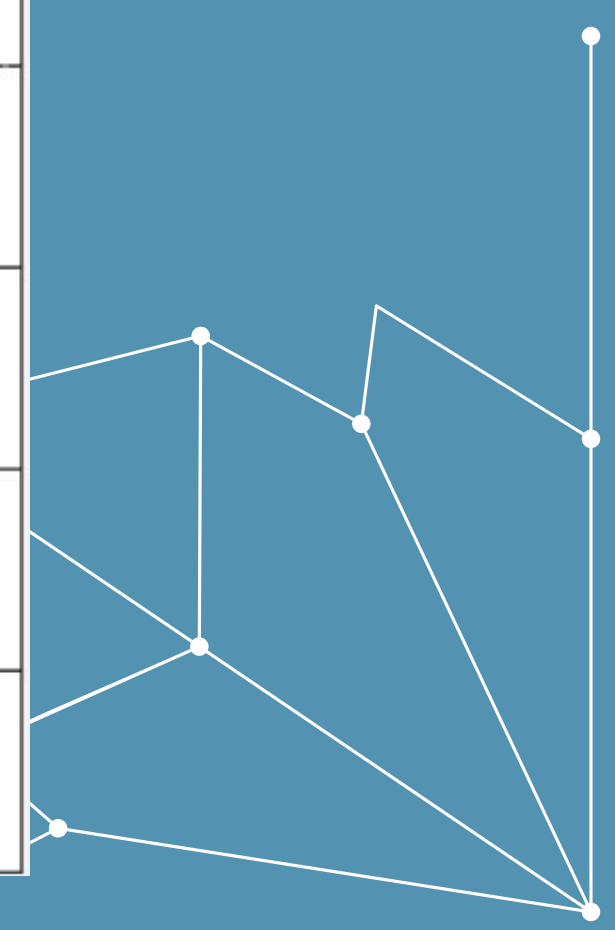
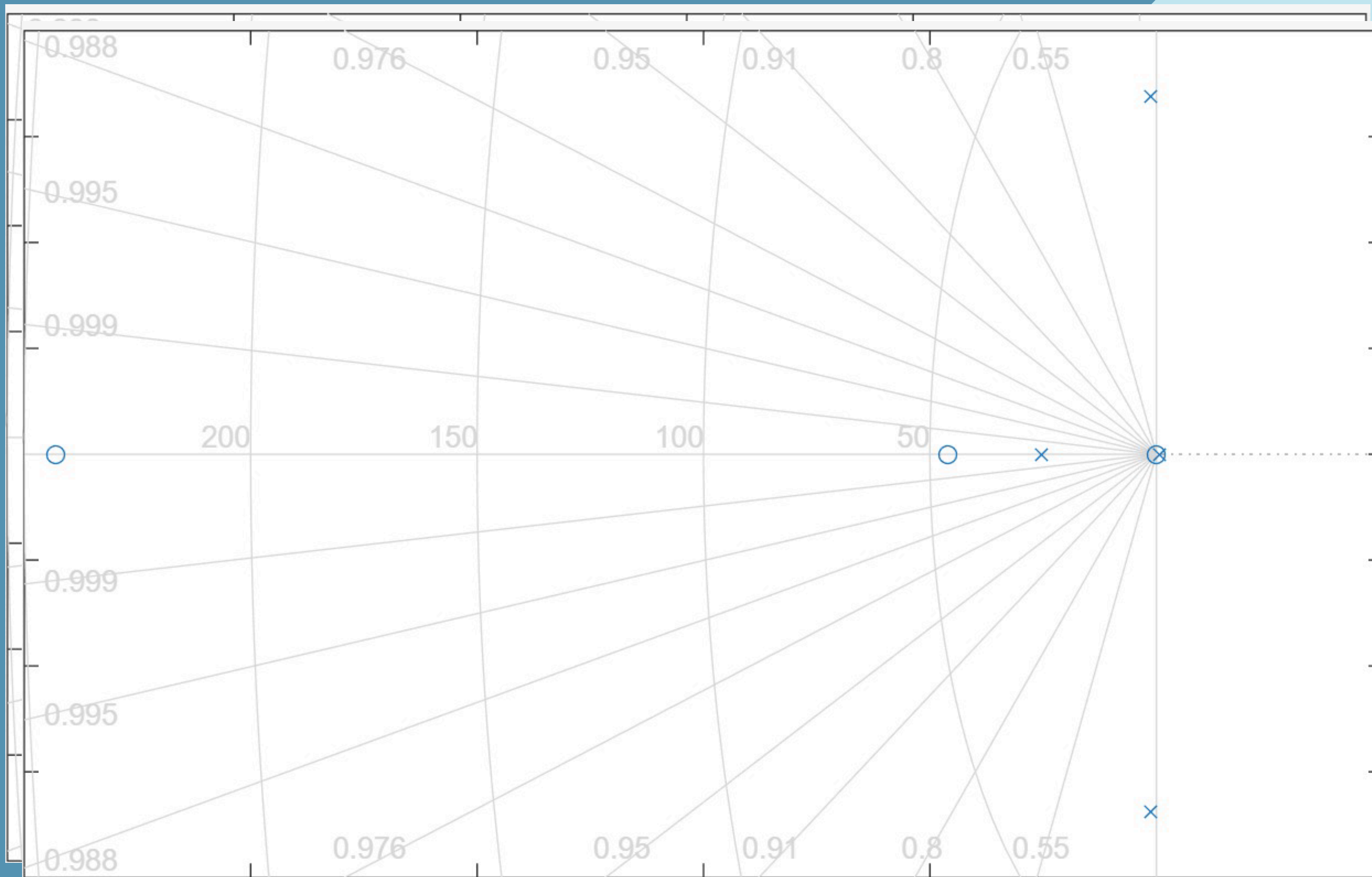
```
-1.2738 +33.8206i  
-1.2738 -33.8206i  
-25.2436 + 0.0000i  
0.8055 + 0.0000i
```

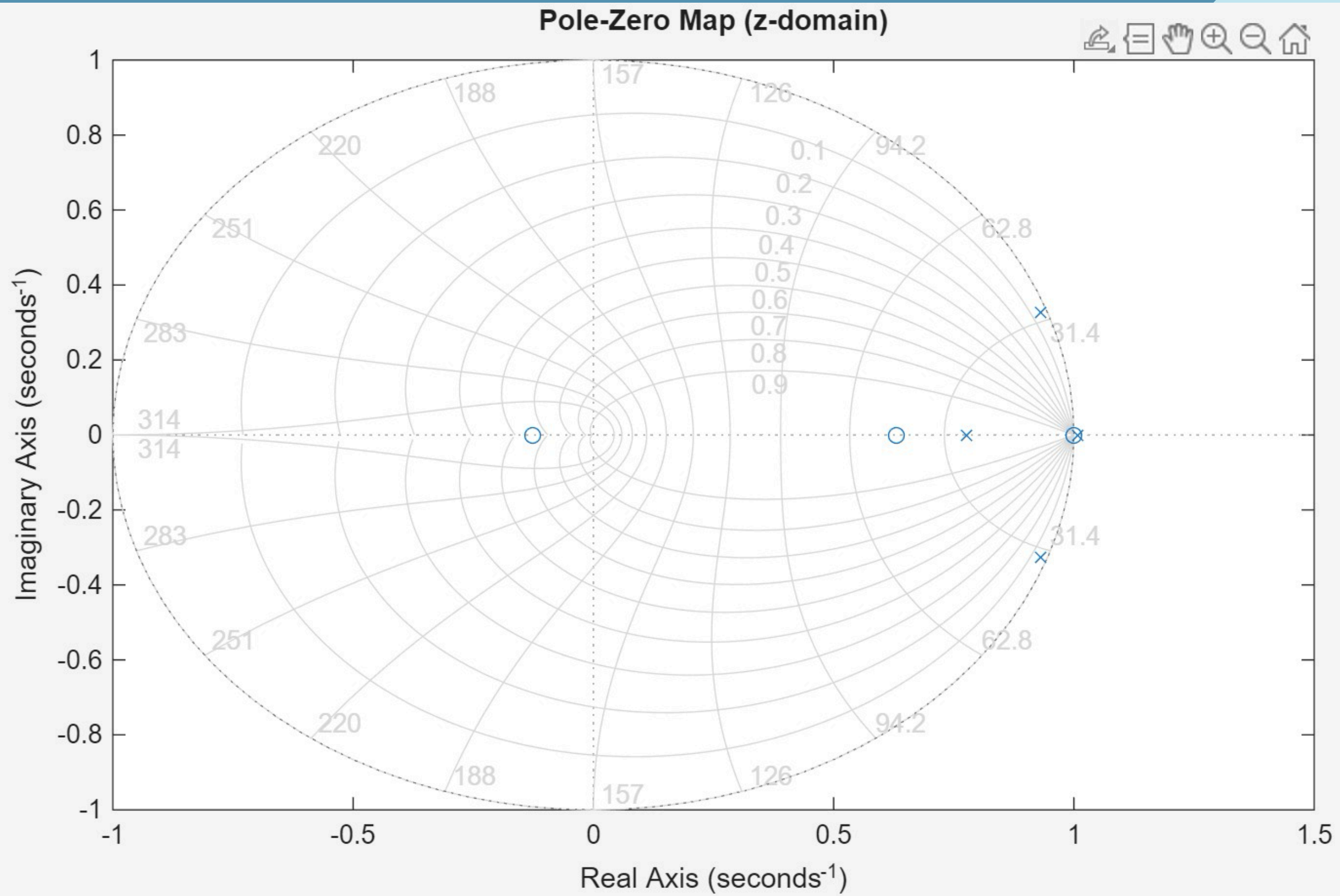
Open-Loop Poles:

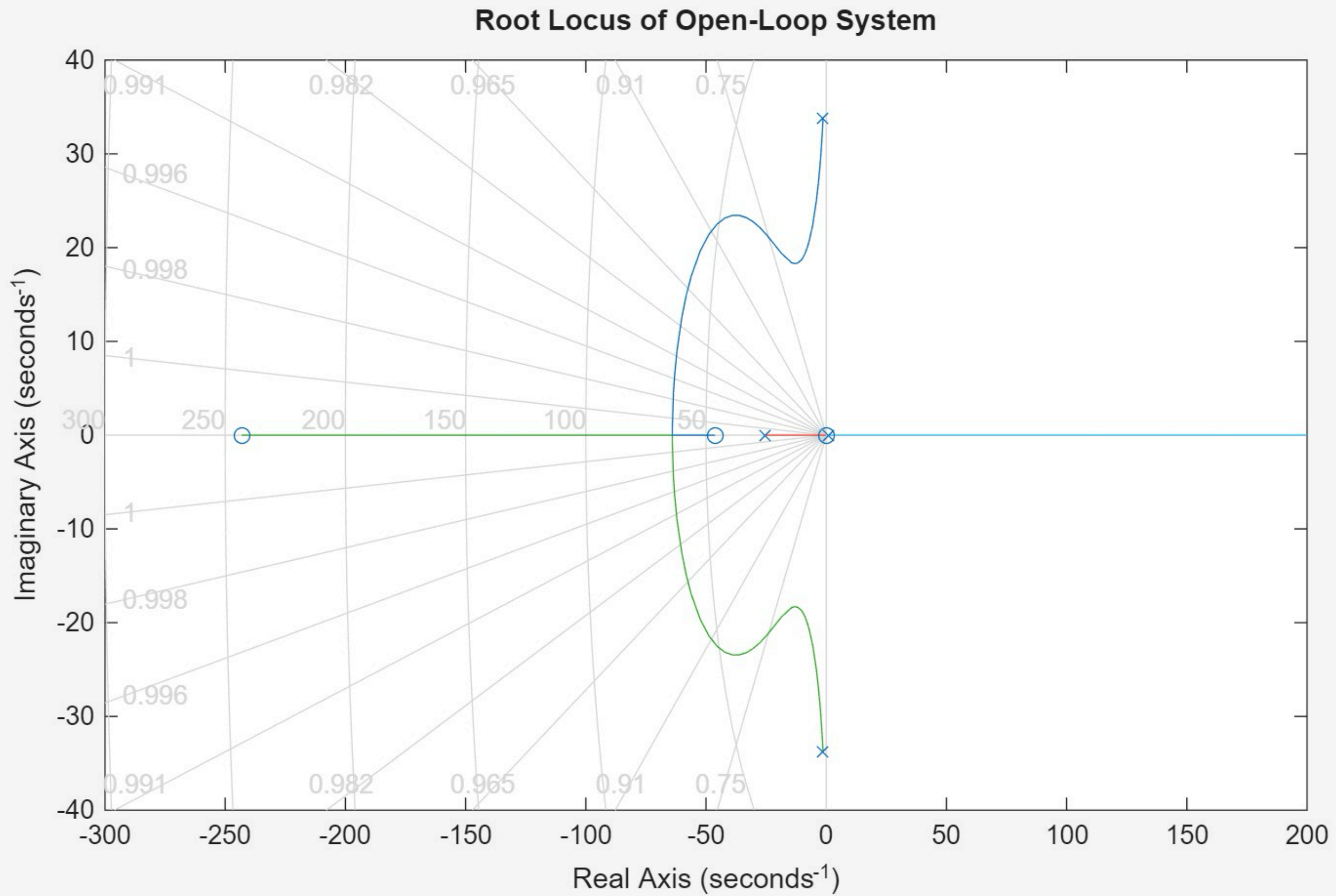
```
ans =
```

```
-1.3081 +33.7910i  
-1.3081 -33.7910i  
-25.2527 + 0.0000i  
0.8029 + 0.0000i
```









System 2(AN INCLINE IN MOTOR POSITION)

With K

G_cl_s =

$$\frac{-0.4424 s^3 + 73.77 s^2 - 531.8 s + 6.603}{s^4 - 17.49 s^3 + 1627 s^2 - 2.805e04 s - 2384}$$

Without K

G_cl_s2 =

$$\frac{-0.4424 s^3 + 73.77 s^2 - 531.8 s + 6.603}{s^4 - 18.99 s^3 + 1629 s^2 - 3.046e04 s - 2589}$$

Continuous-time transfer function.

pole(G_cl_s)
pole(G_cl_s2)

ans with K = $0.1118 + 40.3101i$

$0.1118 - 40.3101i$

$17.3489 + 0.00000i$

$-0.0846 + 0.00000i$

ans without K = $0.1291 + 40.3237i$

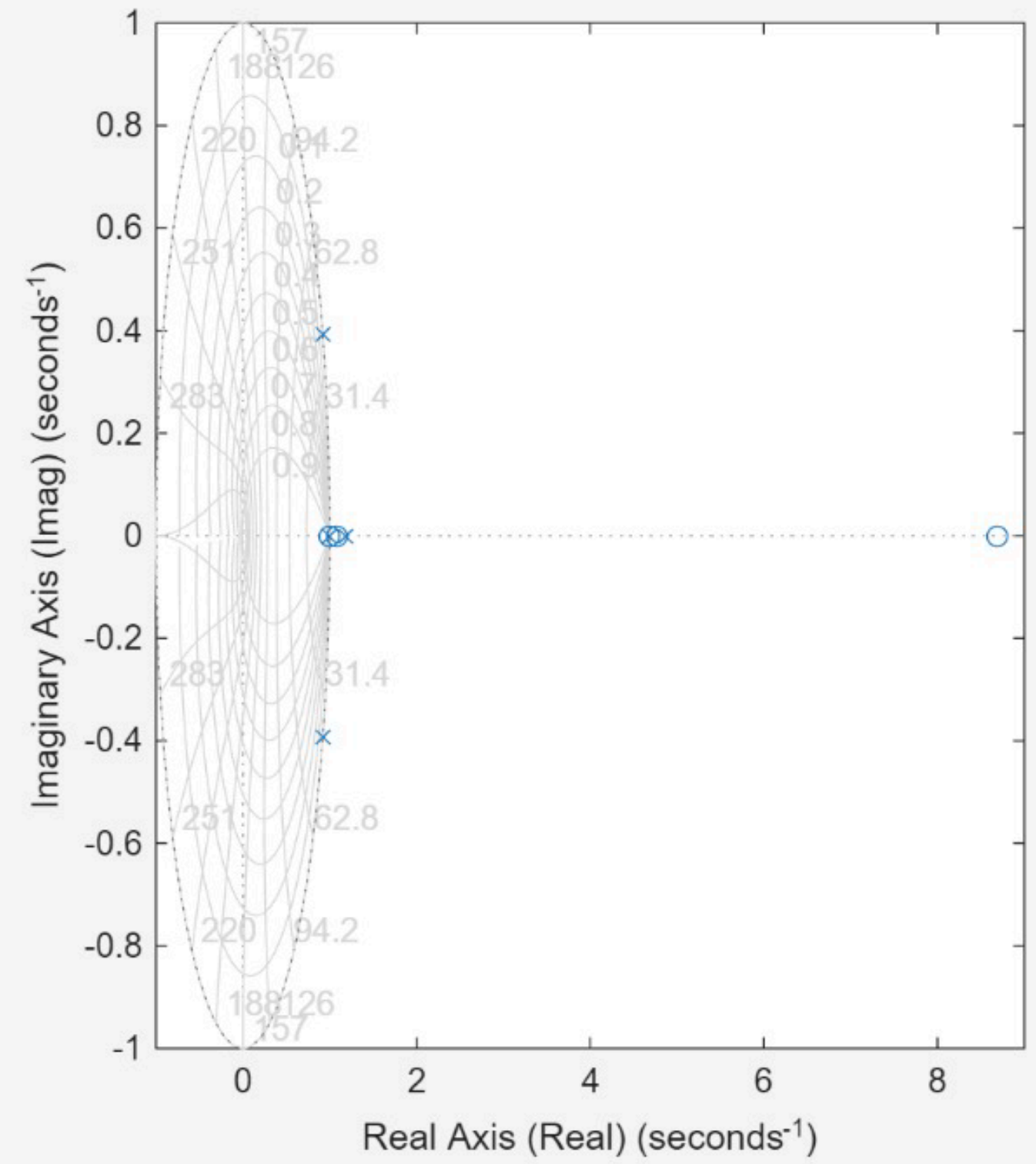
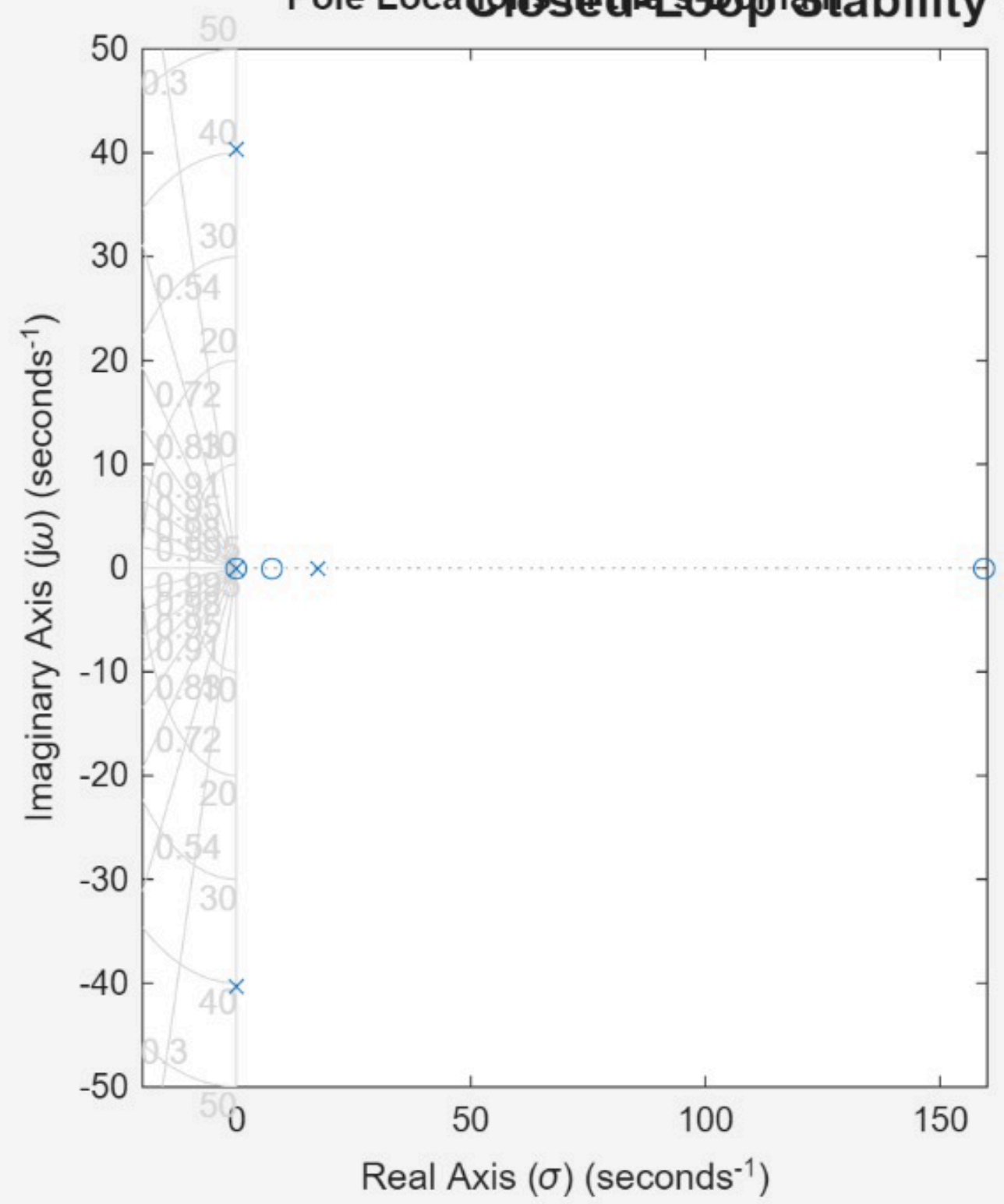
$0.1291 - 40.3237i$

$18.8200 + 0.00000i$

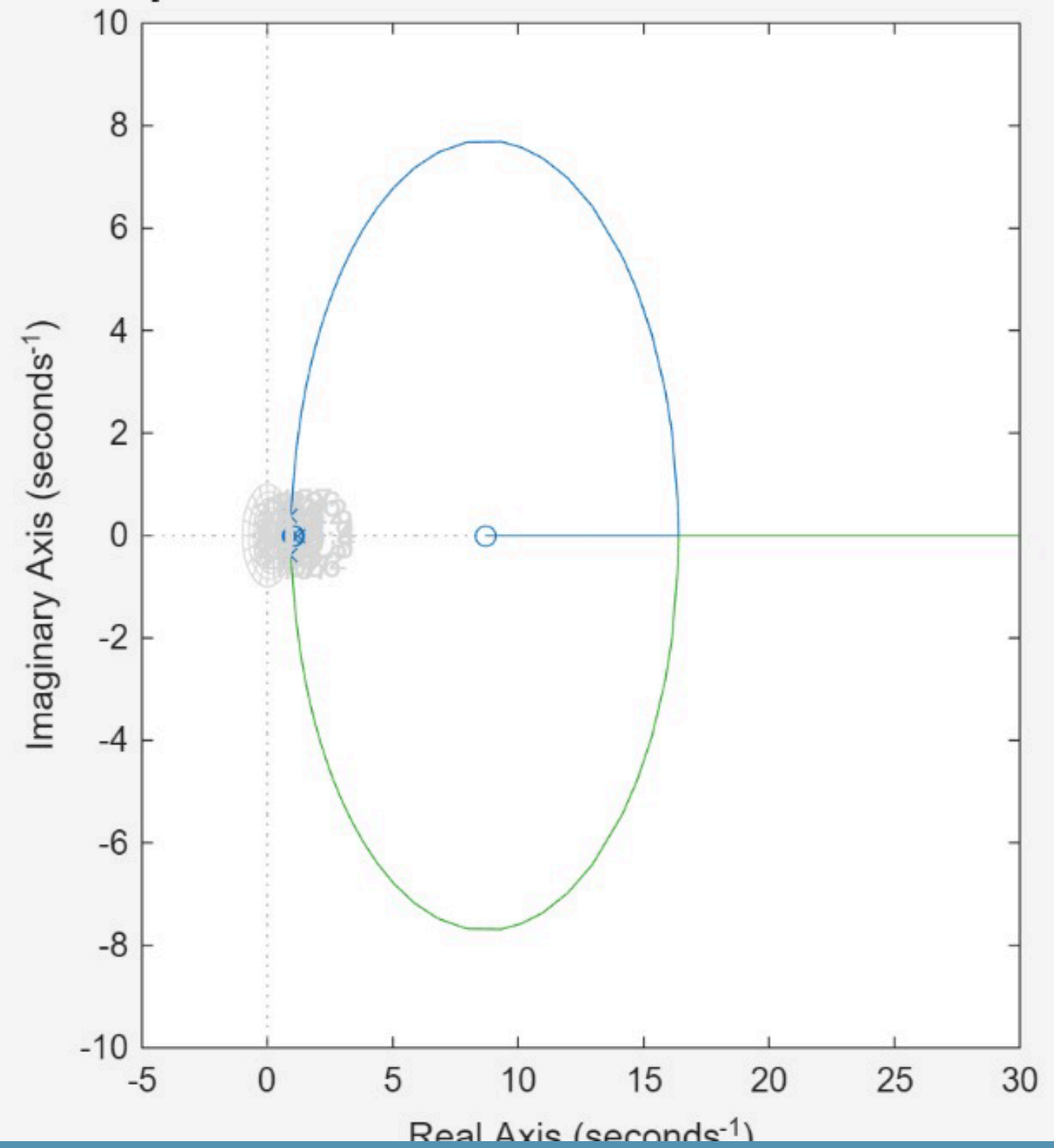
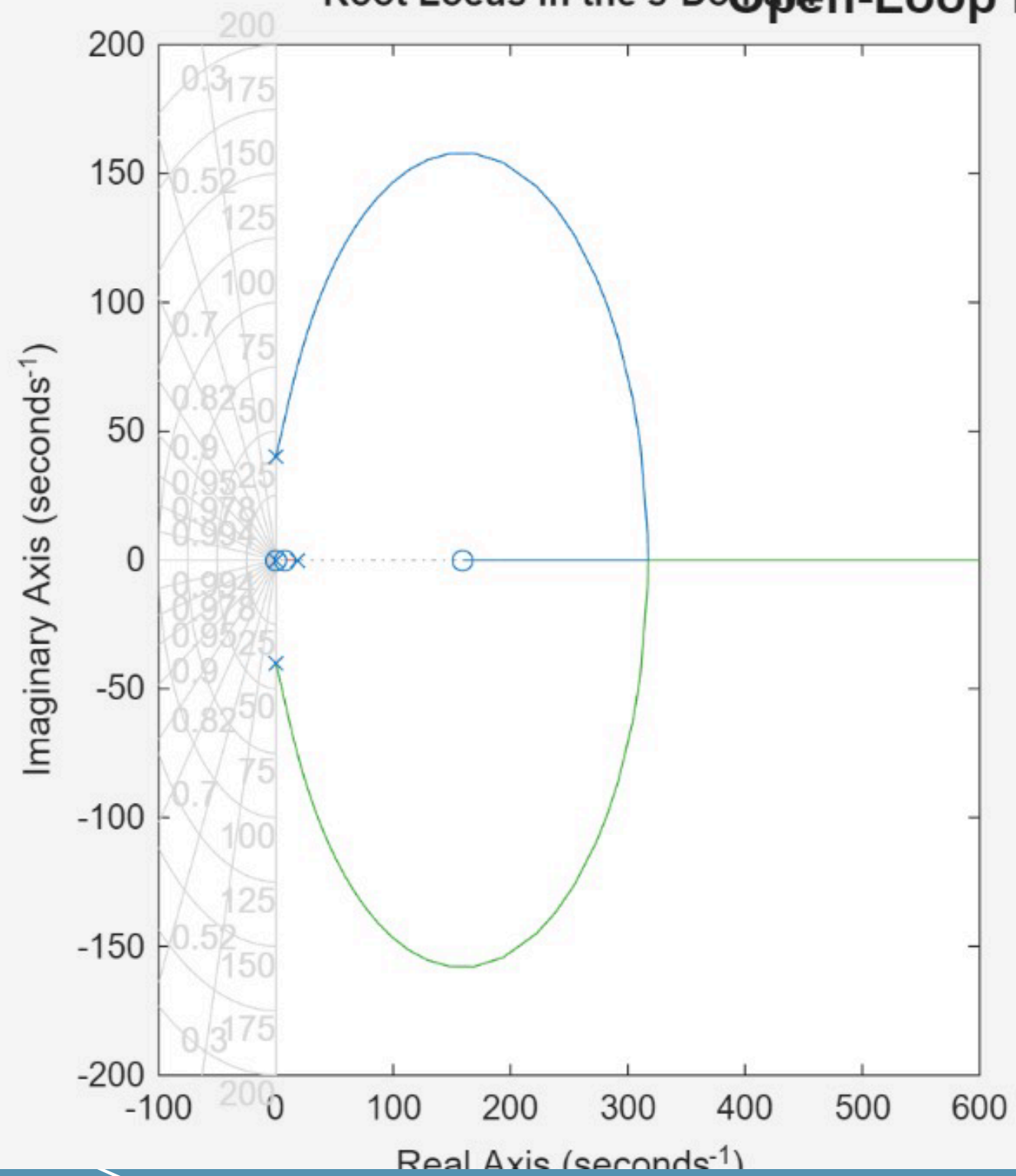
$-0.0846 + 0.00000i$



Closed-Loop Stability Comparison: s-Domain vs z-Domain

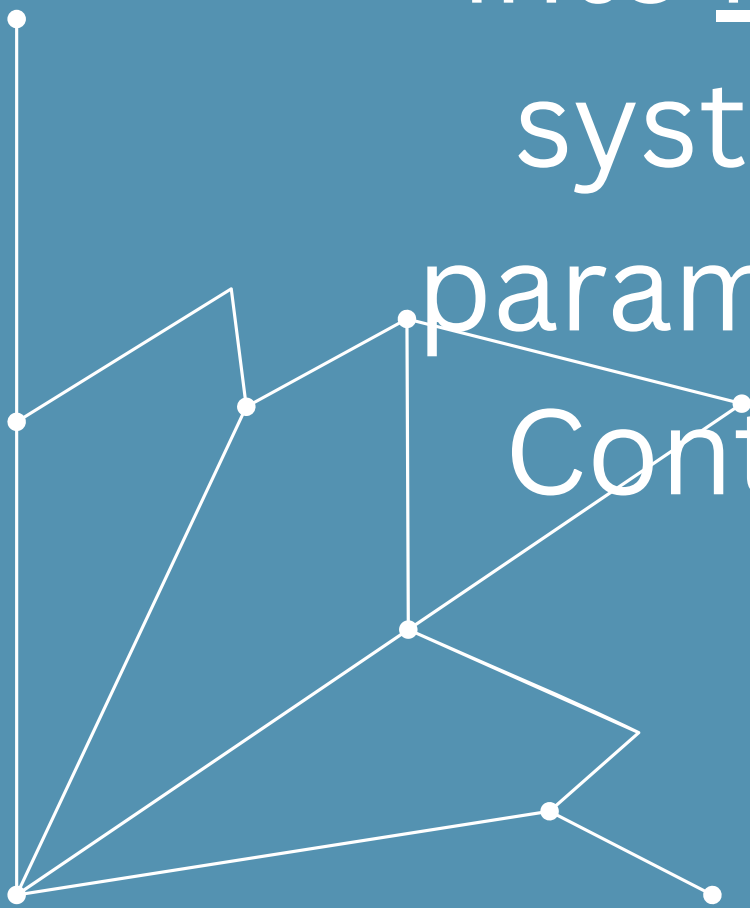


Open-Loop Root Locus Comparison



Here's a concise summary:

The self-balancing robot is a highly sensitive system that first requires an initial control code to achieve basic balance and collect sensor readings. These readings are then captured using CoolTerm and saved as a text file. After that, the data is imported into MATLAB for system identification, where key system parameters are extracted. Finally, these parameters are used to refine and improve the PID Controller for better stability and performance.



challenges

1. surface on which wheels rotate (static friction)
2. The tilt of the motor, as a motor, is not straight
3. The 3rd plate may include a non-straight MPU setpoint difference
4. The maximum tilt that can be handled is a small angle approximation

